

Corrective Action Report — CAR-2025-002

Field	Value
Document ID	CAR-2025-002
Issue Date	2025-08-05
Author	J. Okafor, Reliability Engineer III; reviewed by D. Pham, Compliance Lead
CAR Status	Pending Engineering Review (since 2025-03-18)
Subject	Siemens Gamesa OEM Manual Rev. 4.2 §3.2.1 Vibration Replacement Thresholds Conflict with Cascadia Standard MS-2025-03 §5.2
Affected Region / Site / Asset	CAISO and NW-PowerPool / CAISO-WIND-026, NW-PowerPool-WIND-024 / WIN-10130, WIN-10131, WIN-10011, WIN-10054
Severity	High
Owner	E. Linhart, Principal Engineer — Rotating Equipment
Original Target Close Date	2025-06-30 (missed)
Revised Target Close Date	2026-02-28
Distribution	Engineering Standards Committee; VP Operations; CAISO Plant Manager; NW-PowerPool Plant Manager; Procurement; Warranty Administration

1. Finding

Siemens Gamesa Renewable Energy OEM Service Manual **Rev. 4.2** (document reference **VTM-SIEMENS-GAMESA-4.2**), §3.2.1 "Drivetrain Vibration Limits," prescribes mandatory component replacement (gearbox + high-speed-shaft bearing assembly) when measured nacelle-floor RMS velocity exceeds **4.5 mm/s** averaged across a 10-minute window at rated power, per ISO 10816-21:2015 Class IV. The internally revised **Cascadia Standard MS-2025-03 §5.2**, issued February 2025, raises the action threshold to **6.5 mm/s** with a mandatory refurbishment-first protocol (bearing replacement and oil flush) before any gearbox replacement decision.

This is a **30% relative threshold gap** and a different action class (replace-on-trip vs. refurbish-first). The result is that field engineering teams, when invoking Siemens Gamesa under OEM warranty, are receiving replacement recommendations that Cascadia Engineering would not authorize under MS-2025-03.

Management of Change request **MOC-2025-022** was submitted on 2025-03-18 to formally override the OEM threshold for non-warranty operating decisions. The MOC has been in "Pending Engineering Review" status for 14 months as of issue date of this CAR.

2. Evidence

2.1 Threshold-driven replacement recommendations (FY2025)

Asset	Outage Ref	Date	Measured RMS Velocity	OEM (4.5 mm/s) Action	MS-2025-03 (6.5 mm/s) Action	Decision Taken
WIN-10130	OUT-2025-0079	2025-02-11	5.1 mm/s	Replace gearbox	Refurbish bearing	Replaced (OEM warranty pressure) — \$1.2M
WIN-10131	OUT-2025-0570	2025-08-28	5.8 mm/s	Replace gearbox	Refurbish bearing	Disputed; replacement not executed pending MOC outcome
WIN-10011	OUT-2025-0320	2025-07-08	5.4 mm/s	Replace gearbox	Refurbish bearing	Refurbished by CPS — \$214K
WIN-10054	OUT-2025-0114	2025-04-22	6.2 mm/s	Replace gearbox	Refurbish bearing	Replaced (OEM warranty pressure) — \$1.2M

Two replacements totalling **\$2.4M** were executed in FY2025 on the basis of the OEM threshold; under MS-2025-03 these would have remained in the refurbish-first protocol at ~\$214K each, for an avoidable cost differential of approximately **\$1.97M**.

2.2 Warranty interaction

Siemens Gamesa has stated, in correspondence dated 2025-04-09 (filed VTM-SIEMENS-GAMESA-4.2 supplemental), that "operation above the Rev. 4.2 §3.2.1 limits voids the drivetrain extended warranty for the asset for the remainder of the warranty period." Warranty Administration has interpreted this as binding. Engineering disagrees.

2.3 Repeat-failure correlation

WIN-10131 has 8 forced outages YTD; WIN-10011 has 12. The OEM "replace at trip" rule has not produced a step-change reduction in outage rate on assets where it has been executed (WIN-10130 has had 14 forced outages in FY2025, including post-gearbox-replacement events — see OIR-2025-006).

3. Root Cause Hypothesis

- The Siemens Gamesa Rev. 4.2 thresholds are calibrated to ISO 10816-21:2015 Class IV "alarm" levels, which are designed for protection of individual asset OEM components rather than fleet-level reliability optimization. Application of Class IV alarm-as-action policy is conservative from the OEM's warranty perspective but financially punitive for the operator.
- Cascadia Standard MS-2025-03 was revised in February 2025 to align with IEC 61400-25-6 condition-monitoring guidance and internal vibration-trending data from MISO-WIND-038 (the Vestas baseline fleet — see VTM-VESTAS-2025 §7), which shows successful refurbishment outcomes at 6.0–6.8 mm/s RMS.
- MOC-2025-022 has stalled because closure requires concurrence from Warranty Administration, which has not been able to obtain written acknowledgment from Siemens Gamesa that an override would not void warranty.

4. Conflict With Existing Guidance

This finding is, at its core, a documented conflict between:

1. **VTM-SIEMENS-GAMESA-4.2 §3.2.1** (OEM-authored, warranty-binding).
2. **Cascadia Standard MS-2025-03 §5.2** (internal, refurbishment-first).
3. **MOC-2025-022** (pending; would formalize override of #1 in favor of #2).

The conflict is not theoretical. It has produced two replacement decisions on WIN-10130 and WIN-10054 that Engineering would have classified as refurbish-eligible. It is also the procedural reason that the WIN-10131 gearbox replacement quote (filed under EMR-2025-005) remains contested.

5. Recommended Corrective Action

1. Escalate MOC-2025-022 to Engineering Standards Committee for a forced decision within 60 days. The 14-month pendency is itself a control failure.
2. Issue an interim **Engineering Concession Letter (ECL-2025-009)** authorizing the MS-2025-03 threshold for all non-warranty-period assets effective immediately, while MOC-2025-022 closure is pursued.
3. Direct Warranty Administration to obtain a formal written position from Siemens Gamesa on the warranty implications of operation between 4.5 and 6.5 mm/s RMS. If no response is received within 90 days, the absence of objection shall be treated as constructive acceptance.
4. Until MOC closure, all OEM replacement recommendations triggered solely by the §3.2.1 threshold (i.e., measured velocity 4.5–6.5 mm/s) require Engineering co-sign before procurement. No exceptions delegated to site.
5. Retroactively review the WIN-10130 (OUT-2025-0079) and WIN-10054 (OUT-2025-0114) replacement decisions for cost-recovery applicability.

6. Action Owner & Timeline

Sub-action	Owner	Target Date
Escalate MOC-2025-022 to Standards Committee	E. Linhart	2025-09-30 — missed
Issue ECL-2025-009	Engineering Standards Committee	2025-10-31 — missed
Siemens Gamesa formal warranty position	Warranty Admin (L. Reyes)	2025-12-15 — partial response received, ambiguous
Engineering co-sign gate in CMMS	R. Vasquez	2026-01-31 — in development
Retro review of WIN-10130 / WIN-10054	Internal Audit	2026-03-31

7. Implementation Status

Pending Engineering Review. No threshold change has been executed.

ECL-2025-009 is in draft. The Standards Committee has not met on this item since 2025-11-04; the next scheduled review is 2026-06-12. Two additional replacement recommendations have been issued by Siemens Gamesa under the disputed threshold since this CAR was opened.

Cross-references: VTM-SIEMENS-GAMESA-4.2 §3.2.1 (OEM source of conflict); Cascadia Standard MS-2025-03 §5.2 (internal counter-standard); MOC-2025-022 (pending change request); VTM-VESTAS-2025 §7 (comparative baseline data); OIR-2025-006 (WIN-10130 post-replacement outage review); EMR-2025-005 (WIN-10131 gearbox replacement contested invoice); CAR-2025-004 (Apex emergency contract — Apex sub-contracts Siemens Gamesa OEM techs at 1.8× rate); CAR-2025-009 (repeat-failure escalation policy non-compliance).

End of report.